

Machine Learning Fundamentals

Week 2 – Task 2

House Price Prediction Using Linear Regression

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Course: Machine Learning Fundamentals

Organization: Neurofive Solutions

Dataset: California Housing Dataset (Scikit-learn)

Problem Type

Regression

Unlike classification problems that predict categories, regression predicts continuous numerical values.

The objective of this task is to build a Linear Regression model capable of predicting median house prices using selected housing characteristics.

Objectives

This notebook aims to:

- Load a real-world housing dataset.
- Explore the dataset.
- Select the most relevant features.
- Split the dataset into training and testing subsets.
- Train a Linear Regression model.
- Predict house prices.
- Evaluate the model using RMSE and R^2 Score.
- Visualize Actual vs Predicted Prices.
- Interpret the model performance in simple language.

```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Importing the Dataset

The California Housing Dataset is provided by Scikit-learn.

It contains information collected from the California census and is widely used to learn regression techniques.

Each record represents a district and contains several housing-related features along with the median house value.

```
housing = fetch_california_housing(as_frame=True)

df = housing.frame
```

Preview of Dataset

Before training a Machine Learning model, it is important to inspect the dataset to understand its structure and contents.

```
df.head()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467

	Longitude	MedHouseVal
0	-122.23	4.526
1	-122.22	3.585

2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   MedInc                 20640 non-null  float64
1   HouseAge               20640 non-null  float64
2   AveRooms               20640 non-null  float64
3   AveBedrms              20640 non-null  float64
4   Population             20640 non-null  float64
5   AveOccup               20640 non-null  float64
6   Latitude               20640 non-null  float64
7   Longitude              20640 non-null  float64
8   MedHouseVal            20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
```

```
df.describe()
```

	MedInc	HouseAge	AveRooms	AveBedrms
Population \				
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675
std	1.899822	12.585558	2.474173	0.473911
min	0.499900	1.000000	0.846154	0.333333
25%	2.563400	18.000000	4.440716	1.006079
50%	3.534800	29.000000	5.229129	1.048780
75%	4.743250	37.000000	6.052381	1.099526
max	15.000100	52.000000	141.909091	34.066667

	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.070655	35.631861	-119.569704	2.068558
std	10.386050	2.135952	2.003532	1.153956
min	0.692308	32.540000	-124.350000	0.149990
25%	2.429741	33.930000	-121.800000	1.196000

50%	2.818116	34.260000	-118.490000	1.797000
75%	3.282261	37.710000	-118.010000	2.647250
max	1243.333333	41.950000	-114.310000	5.000010

Dataset Summary

The dataset consists of several numerical housing features.

The target variable is **MedHouseVal**, which represents the median house value for each district.

The dataset contains no missing values, making it suitable for immediate model training.

Feature Selection

For this task, five important features are selected based on their expected influence on house prices.

Selected Features:

- MedInc (Median Income)
- HouseAge (Average House Age)
- AveRooms (Average Number of Rooms)
- AveOccup (Average Occupancy)
- Latitude (Location)

```
features = [  
    "MedInc",  
    "HouseAge",  
    "AveRooms",  
    "AveOccup",  
    "Latitude"  
]  
  
X = df[features]  
y = df["MedHouseVal"]
```

Splitting the Dataset

The dataset is divided into two parts:

- Training Set (80%)
- Testing Set (20%)

The training set is used to train the model, while the testing set evaluates how well the model performs on unseen data.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print("Training Samples :", X_train.shape[0])
```

```
print("Testing Samples :", X_test.shape[0])
```

```
Training Samples : 16512
```

```
Testing Samples : 4128
```

Building the Linear Regression Model

Linear Regression is one of the simplest supervised Machine Learning algorithms.

It establishes a relationship between input features and a continuous target variable by fitting the best possible straight line through the data.

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
LinearRegression()
```

Making Predictions

The trained model is now used to predict house prices for the testing dataset.

```
y_pred = model.predict(X_test)
```

```
results = pd.DataFrame({
    "Actual Price": y_test,
    "Predicted Price": y_pred
})
```

```
results.head(10)
```

	Actual Price	Predicted Price
20046	0.47700	1.050383
3024	0.45800	1.536313
15663	5.00001	2.236133

20484	2.18600	2.733562
9814	2.78000	2.052495
13311	1.58700	2.230999
7113	1.98200	2.787868
7668	1.57500	2.233799
18246	3.40000	2.018245
5723	4.46600	4.244371

Model Evaluation

The performance of the Linear Regression model is evaluated using:

- Root Mean Squared Error (RMSE)
- R^2 Score

RMSE measures the average prediction error, while R^2 indicates how well the model explains the variation in house prices.

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))  
r2 = r2_score(y_test, y_pred)  
print("RMSE :", rmse)  
print("R2 Score :", r2)  
RMSE : 0.8049810972472424  
R2 Score : 0.5055020396798212
```

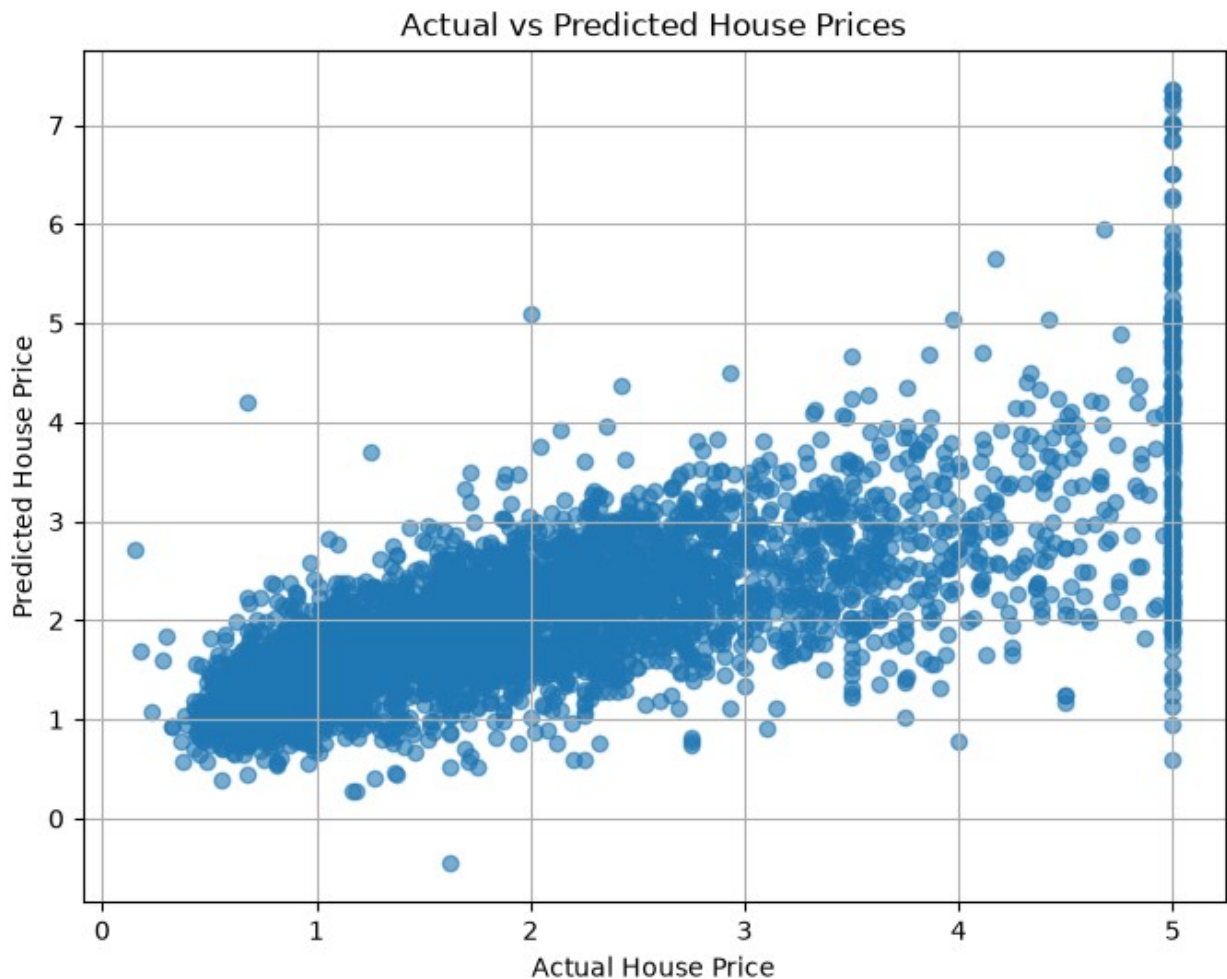
Actual vs Predicted Prices

A scatter plot is used to compare actual house prices with the values predicted by the model.

If the model performs well, the points should appear close to an imaginary diagonal line.

```
plt.figure(figsize=(8,6))  
plt.scatter(y_test, y_pred, alpha=0.6)  
plt.xlabel("Actual House Price")  
plt.ylabel("Predicted House Price")  
plt.title("Actual vs Predicted House Prices")  
plt.grid(True)
```

```
plt.show()
```



What Does the R^2 Score Mean?

The R^2 score tells us how well the model can explain differences in house prices.

For example, if the model achieves an R^2 score of **0.80**, it means that about **80% of the changes in house prices** can be explained by the features used in the model. The remaining **20%** is influenced by other factors that were not included, such as neighborhood quality, nearby facilities, or market conditions.

In simple terms, a higher R^2 score means the model's predictions are generally closer to the actual house prices.

Conclusion

A Linear Regression model was successfully developed to predict house prices using the California Housing Dataset.

Five important housing features were selected for training the model. The model was evaluated using RMSE and R^2 Score, while a scatter plot visually compared actual and predicted prices.

This task demonstrates the complete workflow of a regression problem, from data preparation to model evaluation, and provides a strong foundation for more advanced regression techniques in future Machine Learning projects.